

Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7.	(a)			$2\text{C(s)} + 3\text{H}_2\text{(g)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{C}_2\text{H}_5\text{OH(l)}$ correct formulae (1) balancing and state symbols (1)		2		2		
	(b)			any of following for (1) - if carbon, hydrogen and oxygen are reacted other products (as well as ethanol) would form - the activation energy for the reaction is too high - carbon, hydrogen and oxygen do not easily react with each other under these conditions	1			1		
	(c)	(i)		use of energy / $Q = mc\Delta T$ (1) use of moles (ethanol) = $\frac{\text{mass of ethanol}}{46 / M_r}$ (1) $\Delta_c H = \text{energy} / \text{moles ethanol}$ (1)		3		3		3
		(ii)		ethane is a gas / practical problem based on ethane being a gas			1	1		1
	(d)			Euan: temperature rise / loss in mass very small therefore greater percentage error / less accurate (1) Carys: mass of fuel used after boiling did not result in temperature rise therefore calculated value of $\Delta_c H$ too small (1)			2	2		2